الأسئلة المقالية — 12 عام · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (38 سؤالاً)

LO21 حل المعادلات التربيعية بإكمال المربع · Solve quadratics by completing the square

21.1 LO21 · Ex2

Complete every part of the question shown in the image. Write your working in the space below.

Ex2	.Solve by completing the square حُلّ المعادلة بإكمال المربع.

$$x^2 - 6x + 12 = 19$$

a. Complete all parts of the question.

الأستاذ كامل بشارة 0502837825

21.2 LO21 · 3

Complete every part of the question shown in the image. Write your working in the space below.

3	.Solve by completing the square حُلّ المعادلة بإكمال المربع.

$$x^2 - 12x + 3 = 8$$

a. Complete all parts of the question.

21.3 LO21 · Ex3

Complete every part of the question shown in the image. Write your working in the space below.

Ex 3	.Solve by completing the square	حُلّ المعادلة بإكمال المربع.

$$-2x^2 + 8x - 18 = 0$$

a. Complete all parts of the question.

21.4 LO21 · 3

Complete every part of the question shown in the image. Write your working in the space below.

3	.Solve by completing the square	حُلّ المعادلة بإكمال المربع.

$$3x^2 - 9x - 3 = 21$$

a. Complete all parts of the question.

21.5 LO21 · Ex5

Complete every part of the question shown in the image. Write your working in the space below.

Ex5	.Solve by completing the square بإكمال المربع. حُلّ المعادلة

$$x^2 + 18x - 4 = 0$$

a. Complete all parts of the question.

21.6 LO21 · Ex7

Complete every part of the question shown in the image. Write your working in the space below.

Ex7	.Solve by completing the square	بإكمال المربع.	حُلّ المعادلة

$$4x^2 - 12x - 27 = 0$$

a. Complete all parts of the question.

21.7 LO21 · english-Q2

Solve by completing the square. Show all steps.

$$x^2 - 18x + 81 = 49$$

a. Complete the square

b. Solve and check

21.8 LO21 · english-Q4

Solve by completing the square. Show all steps.

$$4x^2 - 28x + 49 = 64$$

a. Complete the square

b. Solve and check

21.9 LO21 · english-Q5

Solve by completing the square. Show all steps.

$$25x^2 + 20x + 4 = 75$$

a. Complete the square

b. Solve and check

21.10 LO21 · english-Q8

Solve by completing the square. Show all steps.

$$6x^2 - 21x + 9 = 0$$

a. Complete the square

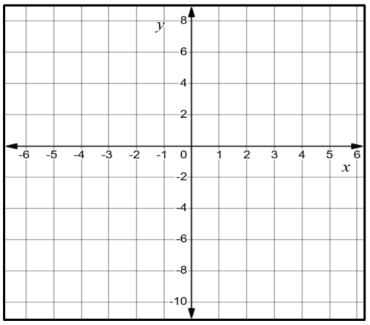
b. Solve and check

22.1 LO22 · Ex2

Complete every part of the question shown in the image. Write your working in the space below.

Ex2 Solve by graphing. حُلّ المتباينة بالتمثيل البياني.

$$x^2 + 2x - 8 < 0$$



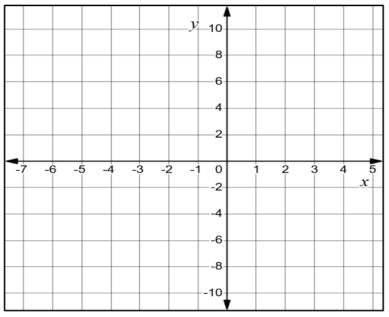
a. Complete all parts of the question.

22.2 LO22 · EX3

Complete every part of the question shown in the image. Write your working in the space below.

EX3 Solve by graphing. حُلّ المتباينة بالتمثيل البياني.

$$2x^2 + 4x - 5 \geq 0$$



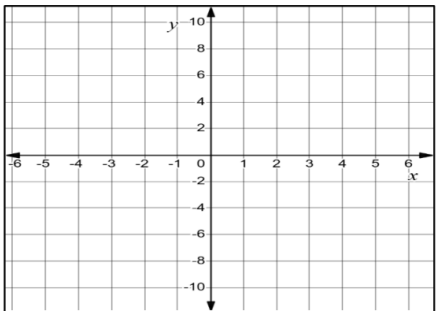
a. Complete all parts of the question.

22.3 LO22 · Ex3

Complete every part of the question shown in the image. Write your working in the space below.

Ex3	Solve by graphing.	حُلّ المتباينة بالتمثيل البياني.

$$x^2 + x - 6 < 0$$



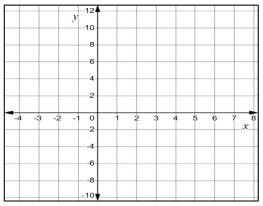
a. Complete all parts of the question.

22.4 LO22 · Ex4

Complete every part of the question shown in the image. Write your working in the space below.

Ex4 Solve by graphing. حل المتباينة بالتمثيل البياني.

$$x^2 - 3x - 4 \geq 0$$



a. Complete all parts of the question.

22.5 LO22 · english-Q11

Solve the inequality by graphing. Write the solution set and show a five-point table.

$$x^2 - 2x - 24 \leq 0$$

a. Five-point table

b. Graph

c. Solution set

22.6 LO22 · english-Q13

Solve the inequality by graphing. Write the solution set and show a five-point table.

$$-x^2 + 3x + 10 < 0$$

a. Five-point table

b. Graph

c. Solution set

22.7 LO22 · english-Q14

Solve the inequality by graphing. Write the solution set and show a five-point table.

$$x^2 - 6x + 2 > 0$$

a. Five-point table

b. Graph

c. Solution set

23.1 LO23 · EX3

Complete every part of the question shown in the image. Write your working in the space below.

EX3			حوّل كل تعبير مما يلي لأبسط صورة.
a)	$(4x^2 - 5x + 6) - (2x^2 + 3x - 1)$	b)	$(6x^2 - 7x + 8) + (-4x^2 + 9x - 5)$

a. Complete all parts of the question.

23.2 LO23 · 32/19

Complete every part of the question shown in the image. Write your working in the space below.

32/19	$(x-y)(x^2+2xy+y^2)$	33	$(a+b)(a^3-3ab-b^2)$

a. Complete all parts of the question.

23.3 LO23 · 38

Complete every part of the question shown in the image. Write your working in the space below.

38	$(x-y)(x+y)(2x+y)$	39/22	$(a+b)(2a+3b)(2x-y)$

a. Complete all parts of the question.

23.4 LO23 · 58/36

Complete every part of the question shown in the image. Write your working in the space below.

58/36	$(2x-2y)^3$	60/38	$(3z-2)^3$

a. Complete all parts of the question.

23.5 LO23 · 5

Complete every part of the question shown in the image. Write your working in the space below.

5	$(6a^2 + 5a + 10) - (4a^2 + 6a + 12)$	10	$(-2f^2 - 3f - 5) + (-2f^2 - 3f + 8)$

a. Complete all parts of the question.

23.6 LO23 · english-Q18

Simplify. Show expansion and collection of like terms.

$$(a+b)(b^3-3ab-b^2)$$

a. Expand

b. Combine like terms

23.7 LO23 · english-Q22

Simplify. Show expansion and collection of like terms.

$(4n-5)^3$

a. Expand

b. Combine like terms

23.8 LO23 · english-Q24

Simplify. Show expansion and collection of like terms.

$$(6x^2 - 7x + 8) + (-4x^2 + 9x - 5)$$

a. Expand

b. Combine like terms

23.9 LO23 · english-Q25

Simplify. Show expansion and collection of like terms.

$$(-x^2 - 3x + 4) - (x^2 + 2x + 5)$$

a. Expand

b. Combine like terms

24.1 LO24 · 9/21

Complete every part of the question shown in the image. Write your working in the space below.

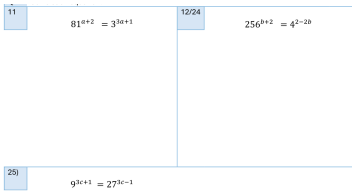
9/21 $8^{4x+2} = 64$

10/22 $5^{x-6} = 125$

a. Complete all parts of the question.

24.2 LO24 · 11

Complete every part of the question shown in the image. Write your working in the space below.



a. Complete all parts of the question.

24.3 LO24 · english-Q30

Solve using a common base. Check whether a solution exists.

$9^{3c+1} = 27^{2c-1}$

a. Common base

b. Solve and check

24.4 LO24 · english-Q31

Solve using a common base. Check whether a solution exists.

$$(1/27)^{2d-2} = 81^{d+4}$$

a. Common base

b. Solve and check

24.5 LO24 ·

Solve and show how you express both sides using a common base.

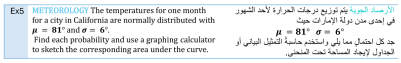
$27^{x-1} = 9^{x+2}$

a. Common base

b. Solve linear equation

25.1 LO25 · Ex5

Complete every part of the question shown in the image. Write your working in the space below.



Ex5 METEOROLOGY The temperatures for one month for a city in California are normally distributed with $\mu = 81°$ and $\sigma = 6°$. Find each probability and use a graphing calculator to sketch the corresponding area under the curve.

الأرصاد الجوية يتم توزيع درجات الحرارة لأحد الشهور في إحدى مدن دولة الإمارات حيث $\mu = 81°$ $\sigma = 6°$. جد كل احتمال مما يلي واستخدم حاسبة التمثيل البياني أو الجداول لإيجاد المساحة تحت المنحنى.

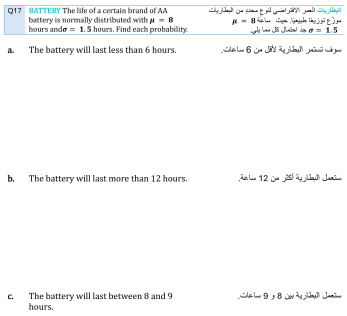
a. $P(70° < X < 90°)$

b. $P(X \geq 95°)$

a. Complete all parts of the question.

25.2 LO25 · Q17

Complete every part of the question shown in the image. Write your working in the space below.



Q17 BATTERY The life of a certain brand of AA battery is normally distributed with $\mu = 8$ hours and $\sigma = 1.5$ hours. Find each probability.

البطاريات العمر الافتراضي لنوع محدد من البطاريات موزع توزيعا طبيعيا، حيث $\mu = 8$ ساعة $\sigma = 1.5$ جد احتمال كل مما يلي:

a. The battery will last less than 6 hours.

سوف تستمر البطارية لأقل من 6 ساعات.

b. The battery will last more than 12 hours.

ستعمل البطارية أكثر من 12 ساعة.

c. The battery will last between 8 and 9 hours.

ستعمل البطارية بين 8 و 9 ساعات.

a. Complete all parts of the question.

25.3 LO25 · Ex8

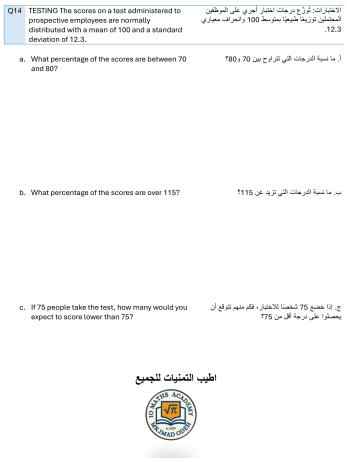
Complete every part of the question shown in the image. Write your working in the space below.



a. Complete all parts of the question.

25.4 LO25 · Q14

Complete every part of the question shown in the image. Write your working in the space below.



a. Complete all parts of the question.

25.5 LO25 · english-Q35

Test scores are normally distributed with mean72 and standard deviation5. Use the standard normal table to find P(X<89). Show standardization.

a. Standardize

b. Normal probability

25.7 LO25 ·

Heights are normally distributed with mean 170 cm and standard deviation 8 cm. Find the height below which 90% of people fall. Use z = 1.28 and round to the nearest tenth.

a. Show all steps and check your answer.

25.7 LO25 ·

Test scores are normal with mean 60 and standard deviation 10. Find the score that cuts off the lowest 5%. Use z = −1.645 and round to the nearest tenth.

a. Show all steps and check your answer.